

Create a Spring Web Project using Maven

pom.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
  https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-parent</artifactId>

    <version>3.2.5</version>

    <relativePath/> <!-- lookup parent from repository -->

  </parent>

  <groupId>com.cognizant</groupId>

  <artifactId>spring-learn</artifactId>

  <version>0.0.1-SNAPSHOT</version>

  <name>spring-learn</name>

  <description>Spring Web Project for Hands-on 1</description>

  <properties>

    <java.version>17</java.version>

  </properties>

  <dependencies>

    <dependency>

      <groupId>org.springframework.boot</groupId>

      <artifactId>spring-boot-starter-web</artifactId>
```

</dependency>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-devtools</artifactId>

<scope>runtime</scope>

<optional>true</optional>

</dependency>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-test</artifactId>

<scope>test</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-maven-plugin</artifactId>

</plugin>

</plugins>

</build>

</project>

src/main/java/com/cognizant/springlearn/SpringLearnApplication.java

java

```
package com.cognizant.springlearn;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class SpringLearnApplication {
```

```
    private static final Logger logger =  
    LoggerFactory.getLogger(SpringLearnApplication.class);
```

```
    public static void main(String[] args) {  
        logger.info("==== Entering main() method of SpringLearnApplication  
====");  
  
        SpringApplication.run(SpringLearnApplication.class, args);  
  
        logger.info("==== SpringLearnApplication started successfully ====");  
    }
```

```
}
```

```
src/main/resources/application.properties
```

```
properties
```

```
# Application name
```

```
spring.application.name=spring-learn
```

```
# Server port (default is 8080)
```

```
server.port=8080
```

```
# Logging level for our package
```

```
logging.level.com.cognizant=DEBUG
```

```
src/test/java/com/cognizant/springlearn/SpringLearnApplicationTests.java
```

```
java
```

```
package com.cognizant.springlearn;
```

```
import org.junit.jupiter.api.Test;
```

```
import org.springframework.boot.test.context.SpringBootTest;
```

```
@SpringBootTest
```

```
class SpringLearnApplicationTests {
```

```
    @Test
```

```
    void contextLoads() {
```

```
    }
```

```
}
```

```
src/main/java/com/cognizant/springlearn/controller/HelloController.java
```

```
java
```

```
package com.cognizant.springlearn.controller;
```

```
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.RestController;
```

```
@RestController
```

```
public class HelloController {
```

```

    @GetMapping("/hello")

    public String sayHello() {

        return "Hello from Spring Learn Application!";

    }

}

```



Hello from Spring Learn Application!

```

WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-07-08T11:29:30.243+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-08T11:29:30.251+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-08T11:29:30.251+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.20]
2026-07-08T11:29:30.281+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-08T11:29:30.281+05:30 INFO 9760 --- [spring-learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed
2026-07-08T11:29:30.490+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-08T11:29:30.508+05:30 INFO 9760 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path
2026-07-08T11:29:30.513+05:30 INFO 9760 --- [spring-learn] [ restartedMain] c.c.spring.learn.SpringLearnApplication : Started SpringLearnApplication in 1.121 seconds (pro
2026-07-08T11:29:30.515+05:30 INFO 9760 --- [spring-learn] [ restartedMain] c.c.spring.learn.SpringLearnApplication : ==== SpringLearnApplication started successfully ==
2026-07-08T11:29:34.225+05:30 INFO 9760 --- [spring-learn] [nio-8080-exec-1] o.apache.tomcat.util.http.parser.Cookie : A cookie header was received [g_state={"i_l":0,"i_ll
Note: further occurrences of this error will be logged at DEBUG level.
2026-07-08T11:29:34.232+05:30 INFO 9760 --- [spring-learn] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherSer
2026-07-08T11:29:34.232+05:30 INFO 9760 --- [spring-learn] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-08T11:29:34.232+05:30 INFO 9760 --- [spring-learn] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms

```

Spring Core – Load Country from Spring Configuration XML

pom.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.cognizant.springlearn</groupId>
  <artifactId>SpringLearnApp</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>

  <properties>
    <maven.compiler.source>1.8</maven.compiler.source>
    <maven.compiler.target>1.8</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <!-- Spring Core -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.30</version>
    </dependency>

    <!-- Logging -->
    <dependency>
```

```
<groupId>org.slf4j</groupId>
<artifactId>slf4j-api</artifactId>
<version>1.7.36</version>
</dependency>
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-log4j12</artifactId>
  <version>1.7.36</version>
</dependency>
<dependency>
  <groupId>log4j</groupId>
  <artifactId>log4j</artifactId>
  <version>1.2.17</version>
</dependency>
</dependencies>
</project>
```

src/main/resources/country.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="country" class="com.cognizant.springlearn.Country">
    <property name="code" value="IN" />
    <property name="name" value="India" />
  </bean>
</beans>
```

</bean>

</beans>

src/main/resources/log4j.properties

properties

log4j.rootLogger=DEBUG, console

log4j.appender.console=org.apache.log4j.ConsoleAppender

log4j.appender.console.layout=org.apache.log4j.PatternLayout

log4j.appender.console.layout.ConversionPattern=%d{yyyy-MM-dd HH:mm:ss} %-5p
%c{1} - %m%n

src/main/java/com/cognizant/springlearn/Country.java

java

package com.cognizant.springlearn;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

public class Country {

private static final Logger LOGGER = LoggerFactory.getLogger(Country.class);

private String code;

private String name;

public Country() {

```
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() {
        LOGGER.debug("Inside getCode() method. Value : {}", code);
        return code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode() method. Value : {}", code);
        this.code = code;
    }

    public String getName() {
        LOGGER.debug("Inside getName() method. Value : {}", name);
        return name;
    }

    public void setName(String name) {
        LOGGER.debug("Inside setName() method. Value : {}", name);
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}
```

src/main/java/com/cognizant/springlearn/SpringLearnApplication.java

java

package com.cognizant.springlearn;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.context.ApplicationContext;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {

private static final Logger LOGGER =
 LoggerFactory.getLogger(SpringLearnApplication.class);

public static void displayCountry() {

ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");

Country country = (Country) context.getBean("country", Country.class);

LOGGER.debug("Country : {}", country.toString());

}

public static void main(String[] args) {

displayCountry();

}

}

```

C:\Users\786as\jdk\openjdk-26.0.1\bin\java.exe ...
2026-07-08 11:39:21 DEBUG ClassPathXmlApplicationContext - Refreshing org.springframework.context.support.ClassPathXmlApplicationContext@7bb11784
2026-07-08 11:39:21 DEBUG XmlBeanDefinitionReader - Loaded 1 bean definitions from class path resource [country.xml]
2026-07-08 11:39:21 DEBUG DefaultListableBeanFactory - Creating shared instance of singleton bean 'country'
2026-07-08 11:39:21 DEBUG Country - Inside Country Constructor.
2026-07-08 11:39:21 DEBUG Country - Inside setCode() method. Value : IN
2026-07-08 11:39:21 DEBUG Country - Inside setName() method. Value : India
2026-07-08 11:39:21 DEBUG SpringLearnApplication - Country : Country [code=IN, name=India]

Process finished with exit code 0

```

pom.xml (updated — add Spring Boot Web Starter)

Since we now need an embedded server (Tomcat) and REST support, switch to **Spring Boot** starters. Replace your existing pom.xml with this:

xml

```

<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <parent>

        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-starter-parent</artifactId>

        <version>2.7.18</version>

        <relativePath/>

    </parent>

    <groupId>com.cognizant.springlearn</groupId>

    <artifactId>SpringLearnApp</artifactId>

    <version>1.0-SNAPSHOT</version>

    <packaging>jar</packaging>

```

```
<properties>

  <java.version>1.8</java.version>

</properties>


<dependencies>

  <!-- Spring Boot Web (includes Spring MVC + embedded Tomcat) -->

  <dependency>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-web</artifactId>

  </dependency>


  <!-- Exclude default logging (Logback) to use Log4j instead -->

  <dependency>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-log4j2</artifactId>

  </dependency>

</dependencies>


<build>

  <plugins>

    <plugin>

      <groupId>org.springframework.boot</groupId>

      <artifactId>spring-boot-maven-plugin</artifactId>

    </plugin>

  </plugins>

</build>

</project>
```

Note: This replaces the earlier plain spring-context + log4j setup with Spring Boot (needed to run an embedded web server on port 8083). Your Country.java/country.xml code from Hands-on 4 still works fine with Boot — no changes needed there.

src/main/resources/application.properties

properties

server.port=8083

logging.level.com.cognizant.springlearn=DEBUG

src/main/java/com/cognizant/springlearn/controller/HelloController.java

java

package com.cognizant.springlearn.controller;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.web.bind.annotation.GetMapping;

import org.springframework.web.bind.annotation.RestController;

@RestController

public class HelloController {

private static final Logger LOGGER = LoggerFactory.getLogger(HelloController.class);

@GetMapping("/hello")

public String sayHello() {

LOGGER.debug("Start :: sayHello()");

String message = "Hello World!!";

LOGGER.debug("End :: sayHello()");

return message;

```
}  
}
```

src/main/java/com/cognizant/springlearn/SpringLearnApplication.java

```
package com.cognizant.springlearn;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
@SpringBootApplication
```

```
public class SpringLearnApplication {
```

```
    private static final Logger LOGGER =  
    LoggerFactory.getLogger(SpringLearnApplication.class);
```

```
    public static void displayCountry() {
```

```
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
```

```
        Country country = (Country) context.getBean("country", Country.class);
```

```
        LOGGER.debug("Country : {}", country.toString());
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        displayCountry();
```

```

SpringApplication.run(SpringLearnApplication.class, args);

}

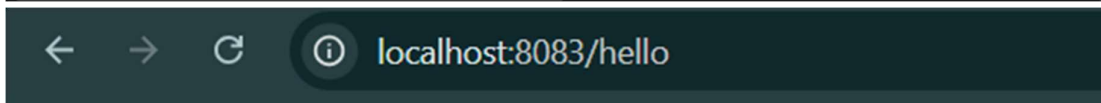
}

```

```

2026-07-08 11:44:30.853 INFO 16372 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8083 (http)
2026-07-08 11:44:30.860 INFO 16372 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-08 11:44:30.860 INFO 16372 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.83]
2026-07-08 11:44:30.927 INFO 16372 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-08 11:44:30.927 INFO 16372 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 547 ms
2026-07-08 11:44:31.119 INFO 16372 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8083 (http) with context path ''
2026-07-08 11:44:31.123 INFO 16372 --- [main] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.998 seconds (JVM running for 1.388)
2026-07-08 11:44:55.003 INFO 16372 --- [nio-8083-exec-1] o.apache.tomcat.util.http.parser.Cookie : A cookie header was received [g_state={"i_l":0,"i_ll":1781980079342,"i_b":
Note: further occurrences of this error will be logged at DEBUG level.
2026-07-08 11:44:55.007 INFO 16372 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-08 11:44:55.007 INFO 16372 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-08 11:44:55.007 INFO 16372 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-08 11:44:55.025 DEBUG 16372 --- [nio-8083-exec-1] c.c.s.controller.HelloController : Start :: sayHello()
2026-07-08 11:44:55.025 DEBUG 16372 --- [nio-8083-exec-1] c.c.s.controller.HelloController : End :: sayHello()

```



Hello World!!

REST - Get country based on country code

File: src/main/resources/country.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="http://www.springframework.org/schema/beans
```

```
    http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="country" class="com.cognizant.spring_learn.Country">
    <property name="code" value="IN" />
    <property name="name" value="India" />
</bean>
```

```
<bean id="countryUS" class="com.cognizant.spring_learn.Country">
    <property name="code" value="US" />
    <property name="name" value="United States" />
</bean>
```

```
<bean id="countryDE" class="com.cognizant.spring_learn.Country">
    <property name="code" value="DE" />
    <property name="name" value="Germany" />
</bean>
```

```
<bean id="countryJP" class="com.cognizant.spring_learn.Country">
    <property name="code" value="JP" />
    <property name="name" value="Japan" />
</bean>
```

```
</beans>
```

File: src/main/java/com/cognizant/spring_learn/service/CountryService.java

```
java
```

```
package com.cognizant.spring_learn.service;
```

```
import java.util.Arrays;
```

```
import java.util.List;
```

```

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;

import com.cognizant.spring_learn.Country;

@Service

public class CountryService {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryService.class);

    public Country getCountry(String code) {

        LOGGER.debug("Start :: getCountry() with code : {}", code);

        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");

        Country country1 = context.getBean("country", Country.class);
        Country country2 = context.getBean("countryUS", Country.class);
        Country country3 = context.getBean("countryDE", Country.class);
        Country country4 = context.getBean("countryJP", Country.class);

        List<Country> countryList = Arrays.asList(country1, country2, country3, country4);

        Country result = countryList.stream()

            .filter(country -> country.getCode().equalsIgnoreCase(code))

            .findFirst()

```

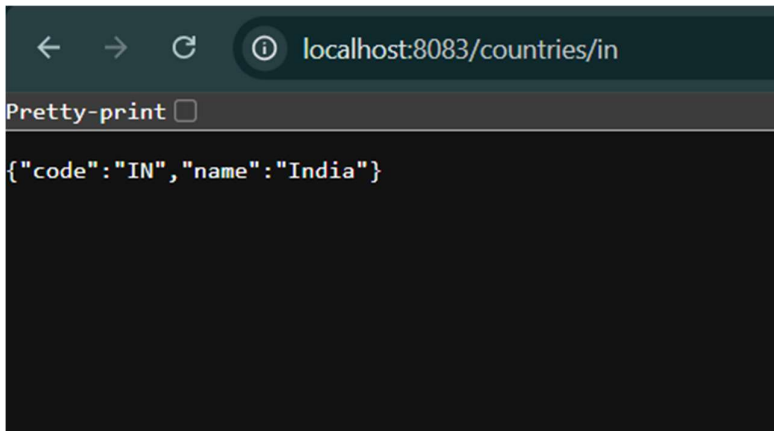
```

        .orElse(null);

    LOGGER.debug("End :: getCountry() returning : {}", result);

    return result;
}
}

```



```

Note: further occurrences of this error will be logged at DEBUG level.
2026-07-08 12:54:52.764 INFO 23252 --- [nio-8083-exec-1] o.a.c.c.{{Tomcat}}.localhost.[] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-08 12:54:52.764 INFO 23252 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-08 12:54:52.764 INFO 23252 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-08 12:54:52.786 DEBUG 23252 --- [nio-8083-exec-1] c.c.s.controller.CountryController : Start :: getCountry() with code : in
2026-07-08 12:54:52.787 DEBUG 23252 --- [nio-8083-exec-1] c.c.s.spring_learn.service.CountryService : Start :: getCountry() with code : in
2026-07-08 12:54:52.911 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode() method. Value : IN
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName() method. Value : India
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode() method. Value : US
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName() method. Value : United States
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode() method. Value : DE
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName() method. Value : Germany
2026-07-08 12:54:52.914 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-08 12:54:52.915 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode() method. Value : JP
2026-07-08 12:54:52.915 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName() method. Value : Japan
2026-07-08 12:54:52.915 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside getCode() method. Value : IN
2026-07-08 12:54:52.915 DEBUG 23252 --- [nio-8083-exec-1] c.c.s.spring_learn.service.CountryService : End :: getCountry() returning : Country [code=IN, name=India]
2026-07-08 12:54:52.915 DEBUG 23252 --- [nio-8083-exec-1] c.c.s.controller.CountryController : End :: getCountry() returning : Country [code=IN, name=India]
2026-07-08 12:54:52.974 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside getCode() method. Value : IN
2026-07-08 12:54:52.974 DEBUG 23252 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside getName() method. Value : India

```

Create authentication service that returns JWT

File: pom.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <parent>

        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-starter-parent</artifactId>

        <version>2.7.18</version>

        <relativePath/>

    </parent>

    <groupId>com.cognizant.springlearn</groupId>

    <artifactId>SpringLearnApp</artifactId>

    <version>1.0-SNAPSHOT</version>

    <packaging>jar</packaging>

    <properties>

        <java.version>1.8</java.version>

    </properties>

    <dependencies>

        <!-- Spring Boot Web (includes Spring MVC + embedded Tomcat) -->

        <dependency>

            <groupId>org.springframework.boot</groupId>

            <artifactId>spring-boot-starter-web</artifactId>
```

</dependency>

<!-- Log4j2 logging -->

<dependency>

 <groupId>org.springframework.boot</groupId>

 <artifactId>spring-boot-starter-log4j2</artifactId>

</dependency>

<!-- Spring Security -->

<dependency>

 <groupId>org.springframework.boot</groupId>

 <artifactId>spring-boot-starter-security</artifactId>

</dependency>

<!-- JWT (jjwt) -->

<dependency>

 <groupId>io.jsonwebtoken</groupId>

 <artifactId>jjwt-api</artifactId>

 <version>0.11.5</version>

</dependency>

<dependency>

 <groupId>io.jsonwebtoken</groupId>

 <artifactId>jjwt-impl</artifactId>

 <version>0.11.5</version>

 <scope>runtime</scope>

</dependency>

<dependency>

 <groupId>io.jsonwebtoken</groupId>

```
<artifactId>jjwt-jackson</artifactId>

<version>0.11.5</version>

<scope>runtime</scope>

</dependency>

</dependencies>


<build>

<plugins>

<plugin>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-maven-plugin</artifactId>

</plugin>

</plugins>

</build>

</project>
```

File: src/main/resources/application.properties

```
properties

server.port=8090

logging.level.com.cognizant.spring_learn=DEBUG


spring.security.user.name=user

spring.security.user.password=pwd
```

File: src/main/java/com/cognizant/spring_learn/util/JwtUtil.java

```
java

package com.cognizant.spring_learn.util;
```

```
import java.security.Key;
import java.util.Date;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.stereotype.Component;

import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.SignatureAlgorithm;
import io.jsonwebtoken.security.Keys;

@Component
public class JwtUtil {

    private static final Logger LOGGER = LoggerFactory.getLogger(JwtUtil.class);

    private final Key key = Keys.secretKeyFor(SignatureAlgorithm.HS256);
    private static final long EXPIRATION_TIME_MS = 1000 * 60 * 20; // 20 minutes

    public String generateToken(String username) {
        LOGGER.debug("Start :: generateToken() for username : {}", username);

        String token = Jwts.builder()
            .setSubject(username)
            .setIssuedAt(new Date(System.currentTimeMillis()))
            .setExpiration(new Date(System.currentTimeMillis() + EXPIRATION_TIME_MS))
            .signWith(key, SignatureAlgorithm.HS256)
            .compact();
    }
}
```

```
        LOGGER.debug("End :: generateToken() generated token : {}", token);

        return token;
    }
}
```

File:

src/main/java/com/cognizant/spring_learn/controller/AuthenticationController.java

```
java

package com.cognizant.spring_learn.controller;

import java.util.HashMap;
import java.util.Map;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.spring_learn.util.JwtUtil;

@RestController

public class AuthenticationController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(AuthenticationController.class);
```

```

@Autowired
private JwtUtil jwtUtil;

@GetMapping("/authenticate")
public Map<String, String> authenticate(Authentication authentication) {

    LOGGER.debug("Start :: authenticate()");

    String username = authentication.getName();
    LOGGER.debug("Authenticated user : {}", username);

    String token = jwtUtil.generateToken(username);

    Map<String, String> response = new HashMap<>();
    response.put("token", token);

    LOGGER.debug("End :: authenticate() returning token for user : {}", username);
    return response;
}
}

```

File: src/main/java/com/cognizant/spring_learn/config/SecurityConfig.java

```

java

package com.cognizant.spring_learn.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;

```

```

import
org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;

import org.springframework.security.web.SecurityFilterChain;

@Configuration
@EnableWebSecurity

public class SecurityConfig {

    @Bean

    public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {

        http

            .csrf().disable()

            .authorizeRequests()

                .antMatchers("/authenticate").authenticated()

                .anyRequest().authenticated()

            .and()

            .httpBasic();

        return http.build();
    }
}

```

```

C:\Users\786as>curl -s -u user:pwd http://localhost:8090/authenticate
{"token":"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ1c2VyIiwiaWF0IjoxNzg5NDk2Mjc5LCJleHAiOiJlE3ODM0OTc0Nz19.3RI01K_h8PPe0ypPtQC5Bf6A
tIJZi1rzGT0BPiXhQ0U"}
C:\Users\786as>

```

